

时间类型	缩写	含义	通俗理解
Access Time (atime)	a	文件内容被读取的时间	「读」文件内容的时间
Modify Time (mtime)	m	文件内容被修改的时间	「改」文件内容的时间
Change Time (ctime)	c	文件元数据被修改的时间	「改」文件属性（权限 / 大小 / 所有者）的时间

操作命令	影响 atime	影响 mtime	影响 ctime	说明
cat file	是	否	否	读取内容, 仅修改访问时间
echo "new" > file	是	是	是	修改内容, 同时改元数据（大小）
chmod 755 file	否	否	是	仅改权限（元数据）
mv file newfile	否	否	是	仅改文件名（元数据）
cp file newfile	新文件 atime = 当前	新文件 mtime = 当前	新文件 ctime = 当前	新建文件
touch file	是	是	是	强制更新所有时间到当前

案例

1: 创建基础测试文件

```
(kali@20232311zfz)-[~/Desktop]
$ touch time_test.txt

(kali@20232311zfz)-[~/Desktop]
$ stat time_test.txt
 文件: time_test.txt
 大小: 0                块: 0                IO 块大小: 4096   regular empty file
设备: 8,1              Inode: 3551213          硬链接: 1
权限: (0664/-rw-rw-r--)  Uid: ( 1000/   kali)   Gid: ( 1000/   kali)
访问时间: 2026-03-16 07:53:43.688466429 -0400
修改时间: 2026-03-16 07:53:43.688466429 -0400
变更时间: 2026-03-16 07:53:43.688466429 -0400
创建时间: 2026-03-16 07:53:43.688466429 -0400
```

2: 修改 atime（仅读取内容）

```
(kali@20232311zfz)-[~/Desktop]
$ cat time_test.txt

(kali@20232311zfz)-[~/Desktop]
$ stat time_test.txt
 文件: time_test.txt
 大小: 0                块: 0                IO 块大小: 4096   regular empty file
设备: 8,1              Inode: 3551213          硬链接: 1
权限: (0664/-rw-rw-r--)  Uid: ( 1000/   kali)   Gid: ( 1000/   kali)
访问时间: 2026-03-16 07:54:01.555955872 -0400
修改时间: 2026-03-16 07:53:43.688466429 -0400
变更时间: 2026-03-16 07:53:43.688466429 -0400
创建时间: 2026-03-16 07:53:43.688466429 -0400
```

3: 修改 mtime + ctime (修改内容)

```
(kali@20232311zfz)-[~/Desktop]
$ echo "test content" > time_test.txt

(kali@20232311zfz)-[~/Desktop]
$ stat time_test.txt
 文件: time_test.txt
 大小: 13          块: 8          IO 块大小: 4096   regular file
设备: 8,1          Inode: 3551213      硬链接: 1
权限: (0664/-rw-rw-r--) Uid: ( 1000/   kali)   Gid: ( 1000/   kali)
访问时间: 2026-03-16 07:54:01.555955872 -0400
修改时间: 2026-03-16 07:54:26.783988196 -0400
变更时间: 2026-03-16 07:54:26.783988196 -0400
创建时间: 2026-03-16 07:53:43.688466429 -0400
```

4: 仅修改 ctime (改权限)

```
(kali@20232311zfz)-[~/Desktop]
$ chmod 600 time_test.txt

(kali@20232311zfz)-[~/Desktop]
$ stat time_test.txt
 文件: time_test.txt
 大小: 13          块: 8          IO 块大小: 4096   regular file
设备: 8,1          Inode: 3551213      硬链接: 1
权限: (0600/-rw-----) Uid: ( 1000/   kali)   Gid: ( 1000/   kali)
访问时间: 2026-03-16 07:54:01.555955872 -0400
修改时间: 2026-03-16 07:54:26.783988196 -0400
变更时间: 2026-03-16 07:54:42.905812845 -0400
创建时间: 2026-03-16 07:53:43.688466429 -0400
```

15 个示例

前置条件:

创建测试目录和文件 (分层结构):

创建不同时间戳的文件 (后续时间查找用) (1 小时前、2 天前、5 分钟前)

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@20232311zfz)-[~/Desktop]
$ mkdir -p find_test/{dir1,dir2,dir3}

(kali@20232311zfz)-[~/Desktop]
$ touch find_test/file1.txt find_test/dir1/file2.log find_test/dir2/file3.sh

(kali@20232311zfz)-[~/Desktop]
$ touch find_test/dir3/file4.pdf find_test/dir3/file5.jpg

(kali@20232311zfz)-[~/Desktop]
$ touch -d "1 hour ago" find_test/old_file1.txt

(kali@20232311zfz)-[~/Desktop]
$ touch -d "2 days ago" find_test/old_file2.log

(kali@20232311zfz)-[~/Desktop]
$ touch -d "5 minutes ago" find_test/recent_file.txt
```

其中 10 个普通 find 命令

1: 查找桌面下所有文件/目录（递归）

```
(kali@20232311zfz)-[~/Desktop]
$ find ~/Desktop
/home/kali/Desktop
/home/kali/Desktop/time_test.txt
/home/kali/Desktop/find_test
/home/kali/Desktop/find_test/dir1
/home/kali/Desktop/find_test/dir1/file2.log
/home/kali/Desktop/find_test/old_file1.txt
/home/kali/Desktop/find_test/dir2
/home/kali/Desktop/find_test/dir2/file3.sh
/home/kali/Desktop/find_test/file1.txt
/home/kali/Desktop/find_test/old_file2.log
/home/kali/Desktop/find_test/dir3
/home/kali/Desktop/find_test/dir3/file4.pdf
/home/kali/Desktop/find_test/dir3/file5.jpg
/home/kali/Desktop/find_test/recent_file.txt
```

2: 仅查找文件（排除目录）

```
(kali@20232311zfz)-[~/Desktop]
$ find ~/Desktop -type f
/home/kali/Desktop/time_test.txt
/home/kali/Desktop/find_test/dir1/file2.log
/home/kali/Desktop/find_test/old_file1.txt
/home/kali/Desktop/find_test/dir2/file3.sh
/home/kali/Desktop/find_test/file1.txt
/home/kali/Desktop/find_test/old_file2.log
/home/kali/Desktop/find_test/dir3/file4.pdf
/home/kali/Desktop/find_test/dir3/file5.jpg
/home/kali/Desktop/find_test/recent_file.txt
```

3: 仅查找目录

```
(kali@20232311zfz)-[~/Desktop]
$ find ~/Desktop -type d
/home/kali/Desktop
/home/kali/Desktop/find_test
/home/kali/Desktop/find_test/dir1
/home/kali/Desktop/find_test/dir2
/home/kali/Desktop/find_test/dir3
```

4: 按文件名精确查找

```
(kali@20232311zfz)-[~/Desktop]
$ find ~/Desktop -name "file1.txt"
/home/kali/Desktop/find_test/file1.txt
```

5: 按文件名模糊查找（忽略大小写）

```
(kali@20232311zfz)-[~/Desktop]
$ find ~/Desktop -iname "*.TXT"
/home/kali/Desktop/time_test.txt
/home/kali/Desktop/find_test/old_file1.txt
/home/kali/Desktop/find_test/file1.txt
/home/kali/Desktop/find_test/recent_file.txt
```

6: 查找空文件（大小为 0）

```
(kali@20232311zfz)-[~/Desktop]
$ find ~/Desktop -type f -empty
/home/kali/Desktop/find_test/dir1/file2.log
/home/kali/Desktop/find_test/old_file1.txt
/home/kali/Desktop/find_test/dir2/file3.sh
/home/kali/Desktop/find_test/file1.txt
/home/kali/Desktop/find_test/old_file2.log
/home/kali/Desktop/find_test/dir3/file4.pdf
/home/kali/Desktop/find_test/dir3/file5.jpg
/home/kali/Desktop/find_test/recent_file.txt
```

7: 查找大小大于 10 字节的文件（因为写入了"test content"，大小超过 10 字节）

```
(kali@20232311zfz)-[~/Desktop]
$ find ~/Desktop -type f -size +10c
/home/kali/Desktop/time_test.txt
```

8: 按深度限制查找（仅查找桌面一级目录下的文件）

```
(kali@20232311zfz)-[~/Desktop]
$ find ~/Desktop -maxdepth 1 -type f
/home/kali/Desktop/time_test.txt
```

9: 正则匹配文件名+批量重命名 (.log→.bak)

```
(kali@20232311zfz)-[~/Desktop]
$ find ~/Desktop -type f -regex ".*/*.*\.log$" -exec mv {} {}.bak \;

(kali@20232311zfz)-[~/Desktop]
$ find ~/Desktop -name "*.bak"
/home/kali/Desktop/find_test/dir1/file2.log.bak
/home/kali/Desktop/find_test/old_file2.log.bak
```

10: 按权限查找 (600 权限的文件)

```
(kali@20232311zfz)-[~/Desktop]
$ find ~/Desktop -type f -perm 600
/home/kali/Desktop/time_test.txt
```

其中 6 个根据时间查找的例子

1: 查找 5 分钟内访问过的文件 (atime)

```
(kali@20232311zfz)-[~/Desktop]
$ find ~/Desktop -type f -amin -5
/home/kali/Desktop/time_test.txt
```

2: 查找 5 分钟内元数据修改过的文件 (ctime)

```
(kali@20232311zfz)-[~/Desktop]
$ find ~/Desktop -type f -cmin -5
/home/kali/Desktop/time_test.txt
```

3: 组合时间条件 (1 小时内访问且 2 天内修改)

```
(kali@20232311zfz)-[~/Desktop]
$ find ~/Desktop -type f -amin -60 -mtime -2
/home/kali/Desktop/time_test.txt
/home/kali/Desktop/find_test/dir1/file2.log
/home/kali/Desktop/find_test/dir2/file3.sh
/home/kali/Desktop/find_test/file1.txt
/home/kali/Desktop/find_test/dir3/file4.pdf
/home/kali/Desktop/find_test/dir3/file5.jpg
/home/kali/Desktop/find_test/recent_file.txt
```

4: 按时间查找并执行操作（删除 7 天前的空文件）

```
(kali㉿20232311zfz)-[~/Desktop]
$ touch -d "7 days ago" ~/Desktop/expired.txt

(kali㉿20232311zfz)-[~/Desktop]
$ ls
expired.txt  find_test  time_test.txt

(kali㉿20232311zfz)-[~/Desktop]
$ find ~/Desktop -type f -empty -mtime +6 -delete

(kali㉿20232311zfz)-[~/Desktop]
$ ls
find_test  time_test.txt
```

5: 查找访问时间在 10 分钟到 1 小时之间的文件

```
(kali㉿20232311zfz)-[~/Desktop]
$ find ~/Desktop -type f -amin +10 -amin -60
/home/kali/Desktop/time_test.txt
/home/kali/Desktop/find_test/dir1/file2.log.bak
/home/kali/Desktop/find_test/dir2/file3.sh
/home/kali/Desktop/find_test/file1.txt
/home/kali/Desktop/find_test/dir3/file4.pdf
/home/kali/Desktop/find_test/dir3/file5.jpg
/home/kali/Desktop/find_test/recent_file.txt
```

6: 查找近 30 分钟内属性被改的文件（比如改权限/重命名）

```
(kali㉿20232311zfz)-[~/Desktop]
$ find ~/Desktop -type f -cmin -30
/home/kali/Desktop/find_test/dir1/file2.log.bak
/home/kali/Desktop/find_test/old_file2.log.bak
/home/kali/Desktop/find_test/time_range2.txt
/home/kali/Desktop/find_test/time_range1.txt
```